

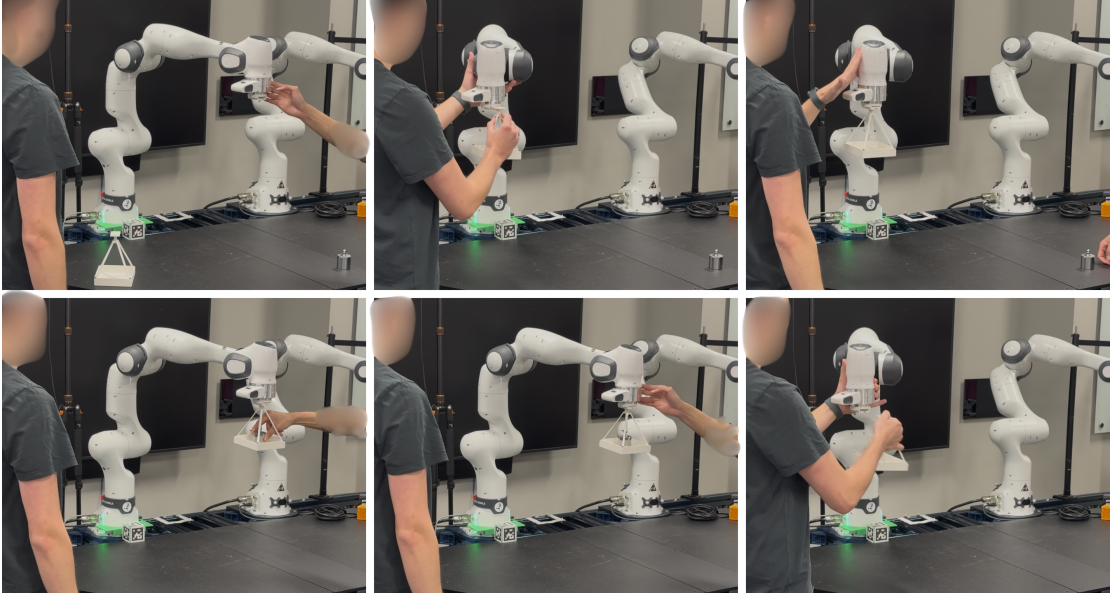


Figure 1: Intent estimation using interaction force estimation. As shown, the framework provides informative estimates of interactions, enabling natural target switching upon detecting human intent.

Hidden Markov Model-based Target Belief filter

To examine solver’s accuracy in estimating interaction forces, we conducted an intent estimation experiment, where the robot utilized $\hat{F}_{\text{int}}$ as the input to a Hidden Markov Model-inspired target belief filter, which estimated to which of the predefined targets the robot should move next by evaluating likelihood function conditioned on the alignment of the interaction force with the distance vector from the robot’s end-effector to the given target. To achieve that, we define n targets for the robot, a belief vector $b_k \in \mathbb{R}^n$, and the transition matrix $M \in \mathbb{R}^{n \times n}$. The state transition is defined as

$$b_{k|k-1} = M^\top b_{k-1} \quad (1)$$

where each entry of M is defined as $M_{ii} = 1 - \alpha$ (stay at target) and $M_{ij} = \frac{\alpha}{n-1}$ (change target), with transition tuning parameter $\alpha = 0.0001$. Then, we define the observation model as the cosine alignment s_k between the interaction force vector F_{int} and the unit vector from the end-effector to a given target u_k as

$$s_k = \frac{F_{\text{int}} \cdot u_k}{\|F_{\text{int}}\| + \epsilon} \quad (2)$$

where $\epsilon = 10^{-6}$. Further, we formulate the likelihood model as

$$L_k = \exp(\beta s_k \|F_{\text{int}}\|) (b_{k-1} + \epsilon)^{\lambda_k} \quad \lambda_k = \exp\left(-\frac{\|F_{\text{int}}\|}{F_0 + \epsilon}\right) \quad (3)$$

with $\beta = 0.001$, where the first term of the likelihood reinforces the target with the strongest force alignment, and the second term reinforces the current target prediction when no interaction forces are present, or they are smaller than $F_0 = 0.3N$. The target belief is then updated as

$$b_k = L_k b_{k|k-1} \quad (4)$$

and the robot proceeds to the target based on its firmest current belief. This formulation allows us to verify the quality of the estimated interaction force. During the evaluation, the framework provided informative estimates of interaction forces, enabling a collaborative human-robot-human transportation task, as shown in Figure 1.